

KENYA JUNIOR SCHOOL EDUCATION ASSESSMENT

GRADE 9 2025



PRETECHNICAL STUDIES

Time: 1 hour 30 Minutes

Name:

School:

Assessment Number:

Signature: Date:

Instructions for Learners

1. Write down your name, grade, school name, and assessment number in the spaces provided.
2. This paper consists of **30 questions**.
3. Ensure you check the question paper to confirm all the pages are printed and no questions are missing.
4. Answer all the questions in the spaces provided or in the answer booklet.

Score Grid

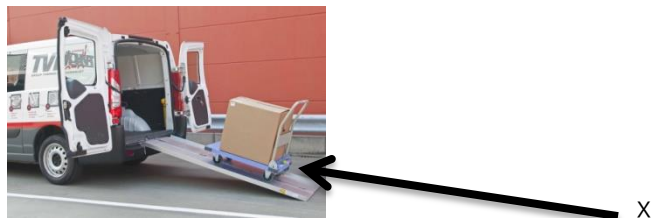
Question	Score	Question	Score	Question	Score
1 – 20		25.		29.	
21.		26.		30.	
22.		27.			
23.		28.		Total Score	
24.				Assessed out of 50	

Assessment Rubrics

Exceeds Expectations (4)	Meets Expectations (3)	Approaching Expectations (2)	Below Expectations (1)
41–50 marks (81–100%)	31–40 marks (61–80%)	21–30 marks (41–60%)	0–20 marks (0–40%)

SECTION A (20 Marks)

1. Grade 9 learners were discussing different types of raised platforms. They displayed the diagram below.



What is the name of the item shown above?

- A. Loading dock B. Work benches
 - C. Catwalks D. Ramps
2. What is the primary purpose of wearing personal protective equipment in a workplace?
- A. To enhance employee comfort
 - B. To comply with fashion guidelines
 - C. To prevent injuries and exposure to hazards
 - D. To improve communication among employees
3. Your sibling requested you to help him discover his talent and you suggested that he should reflect on five things, personal characteristics being among them. Which one of the following is **not** among your recommendations?
- A. Taste B. Ability
 - C. Competence D. Values
4. Grace was teaching her Grade 7 learners about visual programming. She asked her learners to name some advantages of learning visual programming. Which of the following responses was correct?
- A. Requires extensive coding knowledge
 - B. Simplifies the programming process for beginners
 - C. Only useful for professional developers
 - D. Limited to simple applications

5. While driving along a busy highway, Njoki saw the symbol below at a petrol station.



Why was the symbol displayed at the petrol station?

- A. To allow people to warm themselves
 - B. It showed the presence of petrol
 - C. It indicates a cause of fire which should be avoided
 - D. It shows lighting at the petrol station
6. A Grade 9 learner was learning about projection types. Which projection provides a three-dimensional view while maintaining true scale in one dimension?
- A. Isometric projection
 - B. Perspective projection
 - C. Oblique projection
 - D. Axonometric projection
7. Janet went to a computer hardware shop and bought items. Which one of the following is a **pointing device**?
- A. Mouse B. Keyboard
 - C. Flash disk D. Printer
8. A farmer wants to market her new products but she lives in a rural area. Which of the following ICT tools can she use?
- A. Email
 - B. Facebook
 - C. Google
 - D. Poster

9. The diagram below shows a measuring tool.



What is its function?

- A. To measure electric current in a circuit
- B. To smoothen wooden surfaces
- C. To mark and transfer equal measurements
- D. To tighten and loosen bolts and nuts

10. What is the importance of observing safety practices in the workplace?

- A. It boosts employee morale
- B. It increases office gossip
- C. It improves customer satisfaction
- D. It prevents accidents and injuries

11. During a Pre-Studies lesson, Grade 9 learners were discussing the primary angle typically used in oblique projection drawings. Who gave the correct answer?

- A. Dan – 30°
- B. Pauline – 45°
- C. Sande – 60°
- D. Joel – 90°

12. Your class teacher wanted to analyse marks for all learners to calculate the mean grade for the class. Which application software did she use?

- A. PowerPoint
- B. MS Word
- C. MS Excel
- D. MS Edge

13. Your bench vise is not gripping as tightly as it used to. What is the first step to take?

- A. Replace the vise immediately
- B. Clean the jaws and remove any debris

- C. Submerge the vise in water
- D. Apply super glue to the jaws

14. When working on electrical installations, what driving tool is essential for driving screws into junction boxes and electrical panels?

- A. Power drill
- B. Torque wrench
- C. Impact driver
- D. Screwdriver set

15. Which one of the following composite materials is often used in boat building?

- A. Plywood
- B. Fiberglass
- C. Carbon fibre
- D. Aluminium

16. The icon below was displayed by a teacher.



What does it represent?

- A. MS Word
- B. Outlook
- C. Microsoft Edge
- D. MS Excel

17. After surveying their school, Grade 9 learners found a lack of awareness about waste disposal. Which of the following would be the best way to raise awareness among students?

- A. Ignoring issues
- B. Organising a waste management awareness week
- C. Distributing more waste without explanation
- D. Keeping information to themselves

18. Julia, a learner, was asked to define ‘sprite’ as used in visual programming. Which of the following was her response?

- A. A text editor
- B. A type of database
- C. A character or object in a game or animation
- D. A sound file

19. A Grade 9 learner visited a construction site and found workers using concrete. What advantage does concrete provide in construction?

- A. It is biodegradable
- B. It has high compressive strength
- C. It is transparent
- D. It is lightweight

20. While conducting a project, Rina and Wesley identified waste resources in their neighbourhood. Which of the following is a likely significant source of waste in urban areas?

- | | |
|------------------|-------------------|
| A. Local markets | B. National parks |
| C. Rural farms | D. Riverbanks |

SECTION B (30 Marks)

21. Star Academy management has decided to adopt visual programming in their day-to-day learning activities. How will they benefit from using visual programming languages? (3 marks)

- i).....
- ii)
- iii)

22. Mention three potential hazards to personal safety in the environment. (3 marks)

- i).....
- ii)
- iii)

23. The diagram below shows an input device.



(a) What is its name? (1 mark)

.....

(b) Under what category is it grouped? (1 mark)

.....

(c) State its function. (1 mark)

.....

24. Grade 8 learners were asked by their teacher to identify uses of composite materials. Give three responses they gave. (3 marks)

- i)
- ii)
- iii)

25. John recently proposed the installation of the internet in his school. Outline three benefits users would gain from using the internet. (3 marks)

- i)
- ii)
- iii)

26. Angie wants to buy a car for her mother. List three factors she should consider when deciding where to buy the car. (3 marks)

- i)
- ii)
- iii)

27. Njogu is passionate about drawing. She went to a career coach for guidance. Identify three career paths she can choose. (3 marks)

- i).....
- ii)
- iii)

28. Daina, a Grade 8 learner, came across the cutting tool below.



(a) What is its name? (1 mark)

.....

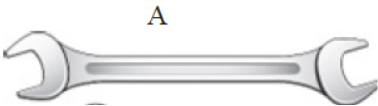
(b) State two ways of ensuring proper care of cutting tools in a work environment. (2 marks)

- i).....
- ii)

29. Outline three safety practices to observe while using holding tools. (3 marks)

- i).....
- ii)
- iii)

30. A Grade 9 learner took the pictures below of driving tools A and B.



B



(a) Identify the tools. (2 marks)

A: B:

(b) State two factors considered while choosing driving tools. (2 marks)

- i).....
- ii)

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ANSWERS

1. D
 2. C
 3. A
 4. B
 5. C
 6. C
 7. A
 8. B
 9. C
 10. D
 11. B
 12. C
 13. B
 14. D
 15. B
 16. A
 17. B
 18. C
 19. B
 20. A
- 21.** - Enhances understanding through use of visual blocks and flowcharts.
- Encourages creativity and problem-solving skills in learners.
 - Makes programming easier for beginners by eliminating the need to memorize syntax.

22. - Slippery floors that can cause falls.

- Exposed electric wires leading to electric shock.
- Sharp objects that may cause injuries.

23. (a) Mouse (b) Pointing devices

24. - Used in making aircraft parts.

- Making sports equipment like tennis rackets.
- Used in construction of buildings and bridges.

25. - Access to online educational resources and research materials.

- Communication through email and video conferencing.
- Enables e-learning and remote learning opportunities.

26. - Reputation of the seller or dealer.

- Price and availability of after-sales service.
- Condition and warranty of the car.

27. - Graphic designer

- Animator
- Architect

28. (a) Planes

(b) - Store in a dry place to prevent rusting.

- Clean and oil the blade after use.

29. - Use the right size tool for the task.

- Keep tools clean and in good condition.
- Ensure a firm grip before applying force.

30. (a) - Wrenches - Screwdriver

(b) - Type of material or fastener to be driven.

- Comfort and safety of the handle.